Work Flow

Epic 1: Problem Definition and Understanding

Story 1

Identify and define the agricultural problem that the crop recommendation system aims to solve.

Story 2

Gather and analyze business requirements to understand project objectives and expected outcomes.

Story 3

Conduct a literature survey to study existing crop recommendation techniques and machine learning approaches.

Story 4

Analyze the social and business impact of implementing an AI-driven crop recommendation solution.

Epic 2: Data Collection and Analysis

Story 1

Download and collect the agricultural dataset from reliable sources.

Story 2

Import the required Python libraries for data analysis, visualization, and machine learning.

Story 3

Read and explore the dataset to understand its structure, features, and target variables.

Story 4

Perform univariate analysis to examine the distribution and characteristics of individual features.

Story 5

Conduct bivariate analysis to identify relationships between agricultural parameters and crop suitability.

Story 6

Perform multivariate analysis to discover patterns and interactions among multiple variables.

Epic 3: Data Preprocessing

Story 1

Check for null values and handle missing data to improve dataset quality.

Story 2

Detect and treat outliers that may affect model performance.

Story 3

Perform feature engineering and prepare relevant features for model training.

Story 4

Split the dataset into training and testing sets for model development and evaluation.

Epic 4: Model Building

Story 1

Apply K-Means Clustering to identify patterns and group similar agricultural conditions.

Story 2

Train a Logistic Regression model to predict suitable crops based on environmental parameters.

Story 3

Evaluate model performance using appropriate metrics and compare results.

Story 4

Save the best-performing model for deployment and generate crop predictions based on user-provided inputs.

Epic 5: Application Building

Story 1

Design and develop HTML pages to create an interactive and user-friendly interface.

Story 2

Build the Python backend using Flask and integrate it with the trained machine learning model.

Story 3

Run, test, and validate the application to ensure accurate crop recommendations and smooth functionality.

Overall Project Flow

Problem Understanding
↓
Collect Dataset
↓
Analyze Dataset
↓
Clean Dataset
↓
Train Machine Learning Model
↓
Evaluate Model
↓
Save Model
↓
Build Flask Backend
↓
Create HTML Frontend
↓
Connect Frontend + Backend + Model
↓
Test Application
↓
Final Project